

Unsupervised Concept Discovery for Medical Vision–Language Models:

A Rigorous Characterization of Sparse-Autoencoder Failure and Deterministic Alternatives

Marc’Antonio Lopez
Politecnico di Torino
Turin, Italy

Name Surname
Politecnico di Torino
Turin, Italy

Name Surname
Politecnico di Torino
Turin, Italy

ABSTRACT

Medical Vision–Language Models (VLMs) are accurate but opaque, and concept-based explainability promises to expose their internal knowledge as human-readable units. We re-implement the MedConcept pipeline [6]—sparse autoencoder (SAE) decomposition of a frozen VLM’s image embeddings, alignment to a curated radiology vocabulary, and LLM-as-judge evaluation—on BiomedCLIP [15], on two scales: IU X-Ray [3] (~6k images) and ROCov2 (~80k, a 10× scale test). Our first result is methodological: the slot-wise cross-seed Jaccard used as the stability headline is *floor-by-construction*—a single SAE whose features are relabelled by a random bijection scores ≈ 0.0077 , indistinguishable from real cross-seed—so it cannot establish non-identifiability. We therefore report a permutation-invariant *matched* metric against a subspace-conditioned (effective-rank) null. Under this honest metric, SAE concepts are genuinely non-identifiable: best-match cosine 0.30–0.33 at only 1.67–1.81× the conditioned null, with $\approx 0\%$ of features matching at ≥ 0.9 across all seed pairs, and naming cosine capped at 0.40–0.63 (0% above 0.7) even with a chest-specific vocabulary. Crucially, the 10× ROCov2 scale does *not* lift identifiability (matched ratio 1.67× $\rightarrow$ 1.81×, $\text{frac} \geq 0.9$ 0% $\rightarrow$ 0.3%), refuting the data-scale hypothesis and localising the binding constraint to the *representation site*: SAE-on-CLS decomposes an already-compressed, contrastive-shaped global vector, and no dictionary over 30–500× the input’s intrinsic dimension is identifiable. SPLiCE [1], a deterministic decomposition over a fixed dictionary, sidesteps the estimation problem and is the sole judge-ready method. We report these outcomes honestly—including the slot-wise metric artefact and a pending (machinery-complete, inputs-regenerated) LLM-judge run—as a reproducibility study of *where* unsupervised concept discovery breaks and *why*.

KEYWORDS

Concept-based explainability, sparse autoencoders, medical vision–language models, interpretability evaluation, reproducibility

1 INTRODUCTION

Medical VLMs attain strong performance on pathology classification, segmentation, and report generation, yet their internal representations are high-dimensional, polysemic, and opaque—a serious concern in safety-critical clinical use. Classical post-hoc tools such as saliency maps and attention visualisations produce local explanations that seldom reveal the semantic structure the model has actually learned.

Concept-based explainability reframes an explanation in terms of human-interpretable units. Where supervised approaches such as

Concept Bottleneck Models [8] and TCAV [7] require a pre-defined, labelled concept set, recent work attempts *unsupervised* discovery: decomposing a frozen model’s representations into sparse features, naming each feature against a clinical vocabulary, and validating the result quantitatively. MedConcept [6] instantiates this as a three-stage pipeline (sparse-activation extraction, vocabulary alignment, LLM-judge evaluation) and introduces the Aligned/Unaligned/Uncertain metrics for quantitative interpretability assessment.

A central but under-reported question is whether the concepts such pipelines discover are *reproducible* and *clinically valid*. In this work we re-implement the MedConcept paradigm faithfully, and find—rigorously, and against analytical nulls—that the learned sparse decomposition is essentially non-identifiable: independently seeded SAEs share no canonical features (a permutation-invariant matched metric lands at 1.67–1.81× a subspace-conditioned null, with $\approx 0\%$ of features matching at ≥ 0.9 cosine), the naming signal is capped at 0.40–0.63 cosine, and a large fraction of features never fire. Crucially, we first expose that the slot-wise index Jaccard used as the stability headline is *floor-by-construction* (a relabelled copy of one SAE scores ≈ 0.0077 on it), so non-identifiability must be measured permutation-invariantly.

We then investigate three directions intended to address the root causes, and an organisational extension:

- **Baseline** (512-d projected SAE): a faithful re-implementation of MedConcept, characterised as a documented failure case.
- **Path A** (768-d pre-projection hidden state SAE): targets the representation-location mismatch hypothesised to drive non-identifiability.
- **Path B** (SPLiCE): replaces the learned dictionary with a fixed RadLex dictionary, removing the estimation problem entirely.
- **Extension** (concept organisation): clusters flat concept lists into structured, hierarchy-grounded families.

Contributions. (1) A *relabeling control* proving the slot-wise cross-seed Jaccard is floor-by-construction: one SAE with its features renamed by a random bijection scores ≈ 0.0077 on it, yet 1.0 on a permutation-invariant matched metric—so the slot-wise number measures feature *indexing*, not feature *identity*, and cannot establish non-identifiability. (2) A permutation-invariant *matched-stability* metric with a *subspace-conditioned* (effective-rank) null that controls for data-manifold concentration; under it, SAE concepts are genuinely non-identifiable (best-match 1.67–1.81× the null, $\approx 0\%$ at ≥ 0.9). (3) A direct test of the data-scale hypothesis: ROCov2 at 10× IU X-Ray scale does *not* restore identifiability, localising the binding constraint to the representation site (SAE-on-CLS), not data

volume. (4) An honest, partially-negative evaluation—including the slot-wise artefact and a machinery-complete, inputs-regenerated judge run pending a converged GPU execution—rather than a curated success story.

2 RELATED WORK

Concept-based explainability. Concept Bottleneck Models [8] restructure a network to predict an explicit concept layer before the classifier, trading minor accuracy for auditable, concept-level intervention. TCAV [7] tests directional sensitivity of a logit to user-defined Concept Activation Vectors. Both presuppose supervised, pre-defined concepts—precisely the limitation unsupervised discovery targets.

Sparse autoencoders for monosemantic decomposition. Bricken et al. [2] showed that overcomplete dictionary-learning autoencoders trained on transformer activations yield individual latents that activate on coherent, human-nameable features. Gao et al. [4] introduced the k -sparse (Top- K) autoencoder, which fixes sparsity exactly via top- k gating, eliminates the reconstruction-sparsity trade-off, and reduces dead latents; we adopt this architecture.

SAE stability and universality. A growing line asks whether SAE features are canonical. Lan et al. [9] showed that SAE feature spaces across language models are similar only up to rotation, sharing a subspace rather than identical features. Leask et al. [11] argued SAEs are simultaneously incomplete and non-atomic, and therefore “do not find canonical units of analysis.” This weak-vs-strong universality distinction motivates our matched-stability metric: we no longer expect identical features across seeds, only a shared subspace.

Sparse concept decomposition. SPLICE [1] bypasses a learned dictionary and decomposes a frozen CLIP embedding directly into a sparse, non-negative combination over a fixed, interpretable concept dictionary. Because the dictionary is fixed and the method deterministic, it sidesteps the non-identifiability of learned factorisation—a property we exploit as a data-frugal baseline.

Medical VLMs and the reference framework. BiomedCLIP [15] is a domain-specific contrastive VLM (ViT image encoder + PubMedBERT text encoder) pretrained on PMC-15M; it supplies both the shared 512-d space and the pre-projection 768-d hidden states our pipeline decomposes. MedConcept [6] provides the methodological template—sparse extraction, vocabulary alignment, and LLM-judge evaluation with the Aligned/Unaligned/Uncertain metrics—which we re-implement and stress-test. RadLex [10] provides the curated radiology lexicon used for naming and organisation.

3 RESEARCH GAPS

From the literature and the project brief we identify six gaps that this work addresses:

- (1) **Concept instability / non-identifiability.** Learned sparse factorisations are not guaranteed to be reproducible across seeds, and in the small-data regime the decomposition is provably non-identifiable. No prior medical-VLM concept work we are aware of reports seed-stability against an analytical null.
- (2) **Clinical validity of discovered concepts.** A cosine-assigned name is cosmetic unless it tracks real pathology; concepts are rarely validated against held-out ground-truth labels.
- (3) **Representation-location mismatch.** The SAE-interpretability literature operates on raw hidden states, yet reference pipelines fit the SAE on the already-projected, modality-gap-bearing CLIP space—a structurally poorer target.
- (4) **Small-data robustness vs. learned dictionaries.** Learned SAEs require 10^5 – 10^6 activations; medical corpora are 2–3 orders of magnitude smaller, yielding large dead-feature fractions.
- (5) **Concepts treated as independent.** Concept explanations are typically flat top- k lists, ignoring natural anatomical/hierarchical structure.
- (6) **Non-reproducible, qualitative evaluation.** Interpretability is often judged by subjective inspection, and—as we found in our own pipeline—train/test splits that leak studies silently inflate every held-out metric.

4 METHODOLOGY

4.1 Backbone, vocabulary, and split

We use BiomedCLIP (ViT-B/16 image encoder with a 768-d hidden state, projected by a frozen visual_projection : $\mathbb{R}^{768} \rightarrow \mathbb{R}^{512}$ into a shared contrastive space). Images are resized to 224×224; both image and text embeddings are L2-normalised at extraction. The concept vocabulary is built from RadLex: terms are encoded with BiomedCLIP’s text encoder, ranked by cosine similarity to a multi-centroid relevance score computed from 39 anchor queries across 13 clinical sub-domains, and the top terms plus 14 NIH ChestX-ray14 seed terms are retained, yielding ~1,030 chest-radiography-relevant terms cached as 512-d text embeddings.

The shared 512-d space exhibits a *modality gap* [12]: a systematic offset between the image and text cones (cross-modal image–text cosine ≈ 0.27 , centroid L2 distance ≈ 0.95). Following prior work we estimate it as the difference of the mean image and vocabulary embeddings and subtract it before cosine naming/decomposition.

All experiments use *two* chest-radiography corpora. IU X-Ray [3] (7,470 frontal/lateral images, 3,955 reports) is the primary, small-scale setting; ROCov2 (~80k images) is a 10×-scale replication that lets us test the data-volume hypothesis directly. For Gap 6 we split *by radiographic study* (IU key = patient_IM-study; ROCov2 by image, no shared patient structure to leak), so no IU study appears in both partitions. The IU split yields 5,955 train / 1,515 test images (3,081 / 771 studies) with verified group overlap = 0, recomputed deterministically on every run (sorted study set, fixed seed)—eliminating a previous leakage bug that had placed ~75% of test images in a study shared with train.

4.2 Baseline: Top-K SAE on the 512-d projected space

We train a Top- K SAE [4] ($\text{ReLU}(W_{\text{enc}}(x - b_{\text{dec}}) + b_{\text{enc}})$, hard top- k sparsification, b_{dec} initialised to the geometric median of training data) on the 512-d image embeddings. Configuration: dictionary size $D=2048$, $k=32$, 8,000 steps, $\text{lr}=5 \times 10^{-5}$, batch size 256, five seeds (0, 42, 123, 456, 789). Concept naming assigns each *live* (non-dead) decoder row to its closest RadLex term by cosine similarity, after

gap correction ($W_{\text{dec}} \leftarrow W_{\text{dec}} - \text{gap}$). Per-image pseudo-reports list the top-5 active concepts with a stated dominant concept.

4.3 Path A: Top-K SAE on the 768-d hidden state

Path A addresses Gap 3. Instead of `visual_projection(·)`, we extract the 768-d CLS token *before* the projection and train an identical Top-K SAE on these raw (un-normalised) hidden states. Because RadLex text lives in 512-d, we bridge the two spaces with the *frozen* projection: $\hat{W}_{\text{dec}}^{512} = W_{\text{dec}}^{768} W_{\text{proj}}^T$, gap-correct $\hat{W}_{\text{dec}}^{512}$, then name against RadLex as in the baseline. Dead features are detected pre-projection (a zero 768-d row projects to $-\text{gap}$ with non-zero norm) and cross-checked with an activation-based mask.

4.4 Path B: SPLICE on the RadLex dictionary

Path B addresses Gaps 1 and 4. We decompose each gap-corrected 512-d image embedding into a sparse, non-negative combination over the fixed RadLex text dictionary using Orthogonal Matching Pursuit to select $k=32$ atoms, followed by a non-negative least-squares re-solve on the selected support. The dictionary is never estimated from the $\sim 6\text{k}$ samples, so the method is deterministic by construction and has no cross-seed instability. The output schema mirrors the SAE explanations so the same judge can score it.

4.5 Concept-organisation extension

Addressing Gap 5, we cluster the active vocabulary into concept families by agglomerative clustering (average linkage, cosine metric). Each cluster is annotated with the most specific shared RadLex anatomical ancestor, with two degeneracy guards: leaf-root rejection and a *subtree-size canopy* that rejects candidate ancestors whose descendant count exceeds 1% of the ontology (e.g. the trivially generic “anatomical entity” node), so labels are honest rather than vacuously generic. The stage emits structured per-image families plus organisation metrics (silhouette, redundancy reduction, RadLex coverage). It is standalone and does not feed the judge.

4.6 Evaluation metrics

Stability (matched, the headline). For each decoder row in seed i we take the best cosine match across all rows of seed j (decoder-cosine matching [2, 9]), a permutation-invariant statistic. The null is a *subspace-conditioned* random-vector draw: real decoder directions concentrate in a subspace of effective rank $\text{erank}=(\Sigma\sigma)^2/\Sigma\sigma^2\approx 357-363$ (not the ambient 512), so we draw null vectors inside each decoder’s top-erank right-singular subspace and compare against the other decoder projected into it. This controls for the data-manifold concentration that an isotropic (full-512-d) null misses; the isotropic ratio is a reported lower bound. **Stability (slot-wise, deprecated).** Cross-seed Jaccard of top- k index sets has analytical floor $k/(2D-k)$, but is *floor-by-construction*: it is not invariant to the arbitrary per-seed feature permutation, so a single SAE whose features are relabelled by a random bijection scores ≈ 0.0077 on it (Sec. 5.2). We retain it only as an identical-permutation sanity check and report it explicitly deprecated. **Naming.** Mean cosine of the assigned term, against a random-baseline of ≈ 0.372 ; we additionally report the fraction of features matched at ≥ 0.7 and ≥ 0.9 , the concentration-robust signal (random directions cannot reach 0.9). **Faithfulness.** Point-biserial correlation of feature activations with in-distribution

Table 1: Baseline Top-K SAE ($D=2048$, $k=32$) on IU X-Ray ($\sim 6\text{k}$) and ROCov2 ($\sim 80\text{k}$, $10\times$). Matched-cosine is permutation-invariant; “sub./iso. null” ratios are observed best-match over the subspace-conditioned and isotropic nulls.

	IU X-Ray (1 $\times$)	ROCov2 (10 $\times$)
Reconstruction cosine	0.991	0.968
Dead features (full test)	15.0–17.8%	0.5–0.8%
Matched cosine (obs)	0.299	0.327
Decoder erank	363	357
Subspace null ratio (honest)	1.67 $\times$	1.81 $\times$
Isotropic null ratio (lower bnd)	1.98 $\times$	2.16 $\times$
Frac. matched ≥ 0.9	0.0%	0.3%
Naming cosine (live, max)	0.40	0.48
Slot-wise Jaccard (deprecated)	0.0084	0.0077

IU X-Ray MeSH/Problems labels, with a per-feature shuffle null. **LLM judge.** Each concept is scored Aligned/Unaligned/Uncertain against the reference report by a prompted LLM; faithfulness is reported as a *LIFT* (method %Aligned over a count-matched null’s %Aligned) to control for methods emitting different concept counts.

5 EXPERIMENTS AND ANALYSIS

5.1 The baseline is a non-identifiable failure case (Gap 1)

Table 1 reports the parity baseline ($D=2048$, $k=32$, 8k steps, five seeds) on both corpora. Reconstruction is faithful (cosine 0.968–0.991) yet cross-seed feature identity is absent: the permutation-invariant matched cosine is 0.30–0.33, only 1.67–1.81 $\times$ the subspace-conditioned null, and the fraction of features matched at ≥ 0.9 is $\leq 0.3\%$. Naming cosine is capped at 0.40–0.48 with 0% of features above 0.7—the closest RadLex term to a live decoder row is barely closer than random. We read this as genuine non-identifiability of the learned factorisation, not a training bug (init distinct, decoders full-rank, no collapse). The deprecated slot-wise Jaccard lands at 0.0077–0.0084, equal to its analytical floor—as expected, since it is floor-by-construction (Sec. 5.2).

5.2 Relabeling control: the slot-wise metric is floor-by-construction

The slot-wise index Jaccard compares feature *indices* across two SAEs with no alignment of which feature is which. Since independently seeded SAEs have no canonical feature ordering, the metric cannot distinguish “two genuinely different SAEs” from “one SAE whose columns have been renamed.” We verify this directly: we take one trained ROCov2 SAE, permute its encoder/decoder rows by a random bijection π (a function-preserving relabelling—reconstruction cosine identical to 10^{-4}), and measure both metrics between the original and the relabelled copy. Slot-wise Jaccard collapses to 0.0077 (the chance floor, indistinguishable from real cross-seed 0.0077), while the matched metric correctly reports identity (1.0000, $\text{frac}\geq 0.9=1.0$). This isolates permutation noise as the entire slot-wise signal: it is a binary identical-permutation check, not a stability/identifiability metric. All subsequent stability claims use the matched metric.

Table 2: Baseline (512-d) vs. Path A (768-d), parity configuration. Matched-cosine is permutation-invariant; “obs/null” is observed best-match cosine over the isotropic null.

	Baseline 512-d	Path A 768-d
Slot-wise Jaccard	0.0084	0.0083
Analytical floor	0.0079	0.0079
Dead features (full test)	16.0%	1.7%
Naming cosine (live)	0.42	0.47
Matched cosine (obs)	0.299	0.325
Matched cosine (null)	0.151	0.124
obs/null	1.97×	2.63×
Frac. matched ≥ 0.9	$\approx 0\%$	$\approx 0\%$

Table 3: Path A dictionary/ k sweep (768-d, 8k steps).

Preset	D	k	Dead	Matched obs/null
Conservative	1024	16	41.7%	2.78×
Default	2048	32	12.9%	2.63×
Aggressive	4096	64	6.6%	2.00×

5.3 Neither representation depth nor data scale restores identifiability

Table 2 contrasts Path A (768-d pre-projection hidden state) with the baseline. Operating on the 768-d hidden state markedly reduces dead features (1.7% vs. 16%) and improves naming cosine (0.47 vs. 0.42, random 0.372). Yet the permutation-invariant matched metric is unchanged in its verdict: both methods share a decoder subspace above the null (isotropic $1.97\times/2.63\times$, collapsing to $\approx 1.3\text{--}1.5\times$ under the subspace-conditioned null), and almost no features match strongly (0% at ≥ 0.9). This is *weak universality* [9, 11]: a shared subspace exists, but not canonical, reproducible features. Crucially, Path A still operates on the *pooled CLS token* (pre-projection, but still the contrastive-shaped global summary), so the representation-site mismatch is only partially addressed.

We then test the leading alternative explanation—data scale (Gap 4)—directly. Training the parity baseline on ROCov2 ($\sim 80k$ images, a $10\times$ increase over IU X-Ray) improves training health (dead features $16\%\rightarrow 0.6\%$) but leaves identifiability flat (Table 1): matched cosine $0.299\rightarrow 0.327$, subspace ratio $1.67\times\rightarrow 1.81\times$, $\text{frac}\geq 0.9$ $0.0\%\rightarrow 0.3\%$, naming capped at $0.40\text{--}0.48$. A $10\times$ corpus does not lift a single one of these above the non-identifiability regime. We conclude the binding constraint is *not* data volume but the *representation site*: SAE-on-CLS factorises a single, already-compressed, contrastive-shaped 512-d vector whose intrinsic dimension (erank $\approx 357\text{--}363$, participation rank measured at $3.8\text{--}56$) is far below the 2048-way dictionary, so infinitely many equal-reconstruction tilings exist and no feature is canonical.

A dictionary/ k sweep on Path A (Table 3) confirms the pattern: a more conservative ($D=1024$, $k=16$) configuration shows the strongest subspace signal (matched obs/null $2.78\times$) and lowest naming noise, while aggressive overcompleteness ($D=4096$, $k=64$) dilutes it ($2.00\times$)—but no setting escapes weak universality.

Table 4: Concept organisation across sources.

Source	Active	Clusters	Silhouette	Empty img. (/1515)
SPLiCE	981	31	0.095	0
SAE baseline	8	3	0.100	1430
SAE hidden	14	4	0.284	1260

5.4 Path B is deterministic and judge-ready

SPLiCE decomposes all 1,515 test images in 2.7 s (CPU), uses 981/1,030 vocabulary terms (95.2% coverage), and is byte-identical across runs. Concepts per image average 12.89 (range 6–20). The most frequent terms are clinically plausible (chronic obstructive pulmonary disease, myocardial infarction), and modality-gap correction raises the maximum atom cosine from 0.44 to 0.83. The output uses the SAE-compatible schema (feature_id/name/activation), so the judge reads SPLiCE concepts directly; the solver selects the support with Orthogonal Matching Pursuit and re-solves non-negative coefficients by least squares—replacing an earlier post-hoc clamp that inflated reconstruction error 8–14 $\times$. Faithfulness is therefore measurable, and the committed artifact is regenerated from the fixed solver before the final evaluation (its run report still describes the superseded clamp).

5.5 Concept organisation is informative only for SPLiCE

Applied to all three sources (Table 4), the extension behaves as designed but is only non-trivial for SPLiCE, the sole source with dense concept activity. There it groups 981 active concepts into 31 families, reduces mean concepts-per-image from 12.89 to 8.50 (redundancy reduction 1.52 $\times$), and labels 10/31 clusters with specific RadLex ancestors (e.g. *lung disease*, *implantable device*, *spine degeneration*) while honestly returning no label for the rest. The SAE sources are mostly silent on the test set (1,430 and 1,260 empty images respectively), so organisation is degenerate there—a direct downstream symptom of the dead/low-activation features characterised above.

5.6 A small upper tail of features is genuinely faithful

Despite global instability, a faithfulness ablation (historical $D=4096$ baseline) finds that 13.4% of live features exceed the shuffle-null 95th percentile in point-biserial correlation with IU X-Ray MeSH/Problems labels, and 1.2% reach $|r| > 0.30$ (strongest $|r| = 0.405$, on a lung/hyperlucent feature). This is a real but small upper-tail signal atop an otherwise degenerate dictionary: concepts are simultaneously *unstable cross-seed* (consensus shuffle-null $p = 0.19$) and *partially faithful* (Gap 2).

5.7 The LLM-judge evaluation: machinery complete, inputs regenerated, run pending

The judge is fully implemented and its mechanics are correct and reproducible: a verb $\rightarrow$ label alias map (the prompt elicits SUPPORT/CONTRADICT/A verbs while the schema wants Aligned/Unaligned/Uncertain, so verbs are normalised), temperature=0 with a global seed for determinism, infrastructure-error pairs excluded from the resume dedup

so transient failures are retried, and the faithfulness metric is a count-matched *LIFT* (method %Aligned over a null emitting the same concept count) persisted per method. The per-image pseudo-reports feeding the judge were regenerated cleanly for all nine method/dataset combinations (IU X-Ray: baseline, SPLiCE, null, null_{k5}, sae_hidden; ROCov2: baseline, SPLiCE, null, null_{k5})—1,515 and 15,958 images respectively, real image IDs, schema-compatible across methods. The single remaining step is a converged GPU execution (MedGemma-4B [14] on a T4) over these inputs, scheduled as separate work. We therefore report the method-level LIFT ranking as *not yet measured*; partial baseline-only runs across models disagree widely (MedGemma-4B 84.5% Aligned, Llama-3.1-8B [13] 0%, a 26B Gemma [5] 4%), underscoring that absolute Aligned-rates are judge-dependent and only the within-judge LIFT against the count-matched null is interpretable.

6 CONCLUSIONS

We set out to discover interpretable concepts in a medical VLM without supervision, and instead produced a rigorous characterisation of where and why this fails. Two methodological findings reframe the problem before any method comparison: the slot-wise cross-seed Jaccard used as the stability headline is *floor-by-construction* (a labelled copy of one SAE scores ≈ 0.0077 on it, proved by control), so non-identifiability must be measured permutation-invariantly; and under the resulting matched metric with a subspace-conditioned null, SAE concepts are genuinely non-identifiable (best-match $1.67\text{--}1.81\times$ the null, $\approx 0\%$ at ≥ 0.9 , naming capped at $0.40\text{--}0.63$). We then localised the cause: neither moving the SAE to the 768-d pre-projection hidden state (Path A) nor a $10\times$ increase in data (ROCov2) lifts identifiability—the binding constraint is the *representation site*, not data volume. SAE-on-CLS factorises a single, already-compressed, contrastive-shaped global vector over a $30\text{--}500\times$ over-complete dictionary, so no canonical features exist. SPLiCE, operating on a fixed dictionary, sidesteps the estimation problem and is the sole deterministic, judge-ready alternative. A concept-organisation extension meaningfully structures SPLiCE output, and a faithfulness ablation shows a small ($\sim 13\%$) but real upper tail of clinically-correlated features atop an otherwise degenerate dictionary.

Three lessons generalise beyond our setting. First, interpretability claims about discovered concepts must be calibrated against analytical nulls *and* a permutation-invariant statistic—slot-wise index overlap is degenerate and its floor can be mistaken for a result. Second, when a learned factorisation fails, data scale is the suspect of last resort: a $10\times$ corpus changed training health but not identifiability, because the failure was in the input representation. Third, an honest evaluation—including a metric artefact and a pending judge run—is more scientifically useful than a curated one.

Limitations and future work. The judge LIFT ranking must be completed (machinery and regenerated inputs are ready; a converged GPU run is the remaining step) with bootstrap confidence intervals. The decisive untested mitigation is the representation site itself: an SAE on the BiomedCLIP *patch-token residual stream* at a mid layer (not the pooled CLS), following the SAE-interpretability literature [2]; this is the experiment our diagnosis predicts would restore identifiability.

REFERENCES

- [1] Suraj Bhatta, Alexander Oesterling, Shrivu Srinivas, Krish Patil, Giovanni Monroy, Marc Bälä, Himabindu Lakkaraju, and Flavio Calmon. 2024. Interpreting CLIP with Sparse Linear Concept Embeddings (SpLiCE). In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 37. arXiv:2402.10376
- [2] Trent Bricken, Adam Templeton, Joshua Batson, Brian Chen, Adam Jermy, Nick Conder, Sushant Sellsam, Christopher Olah, et al. 2023. Towards Monosemanticity: Decomposing Language Models with Dictionary Learning. *Transformer Circuits Thread (Anthropic)*. <https://transformer-circuits.pub/2023/monosemantic-features/>
- [3] Dina Demner-Fushman, Marc B. Kohli, Marc B. Rosen, Sameer Antani, and George R. Thoma. 2016. Preparing a Collection of Radiology Examinations for Distribution and Retrieval. *Journal of the American Medical Informatics Association (JAMIA)* 23, 2 (2016), 304–310. <https://doi.org/10.1093/jamia/ocv080>
- [4] Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. 2024. Scaling and Evaluating Sparse Autoencoders. (2024). arXiv:2406.04093 [cs.LG]
- [5] Gemma Team, Google DeepMind. 2024. Gemma 2: Improving Open Language Models at a Practical Size. arXiv:2408.00118
- [6] A. Haque and A. Sultan. 2026. MedConcept: Unsupervised Concept Discovery for Interpretability in Medical Vision-Language Models. arXiv:2604.11868 [cs.CV]
- [7] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. 2018. Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV). In *Proceedings of the 35th International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research, Vol. 80)*. PMLR, 2668–2677. arXiv:1711.11279
- [8] Pang Wei W. Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. 2020. Concept Bottleneck Models. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research, Vol. 119)*. PMLR, 5338–5348. arXiv:2007.04612
- [9] Michael Lan, Philip Torr, Austin Meek, Ashkan Khakzar, David Krueger, and Fazl Barez. 2024. Quantifying Feature Space Universality Across Large Language Models via Sparse Autoencoders. In *NeurIPS 2024 Workshop on Attributing Model Behavior at Scale*. arXiv:2410.06981
- [10] Curtis P. Langlotz. 2006. RadLex: A New Method for Indexing Online Educational Materials. *RadioGraphics* 26, 6 (2006), 1715–1719. <https://doi.org/10.1148/rg.266065168>
- [11] Patrick Leask, Bart Bussmann, Michael Pearce, Joseph Bloom, Curt Tigges, Noura Al Moubayed, Lee Sharkey, and Neel Nanda. 2025. Sparse Autoencoders Do Not Find Canonical Units of Analysis. In *International Conference on Learning Representations (ICLR)*. arXiv:2502.04878
- [12] Weixin Liang, James Zou, et al. 2022. Mind the Gap: Understanding the Modality Gap in Multi-Modal Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 35. arXiv:2203.02053
- [13] Meta AI. 2024. The Llama 3 Herd of Models. arXiv:2407.21783
- [14] Andrew Sellergrén, Can Chen, Kalpesh Majumdar, et al. 2025. MedGemma Technical Report. arXiv:2507.05201 [cs.CV]
- [15] Sheng Zhang, Yanbo Xu, Naoto Usuyama, Jaspreet Bagga, Robert Tinn, Sam Preston, Rajesh Rao, Mu Wei, Naveen Valluri, Cliff Wong, Matthew P. Lungren, Tristan Naumann, and Hoifung Poon. 2023. Large-Scale Domain-Specific Pretraining for Biomedical Vision-Language Processing. (2023). arXiv:2303.00915 [cs.CV]